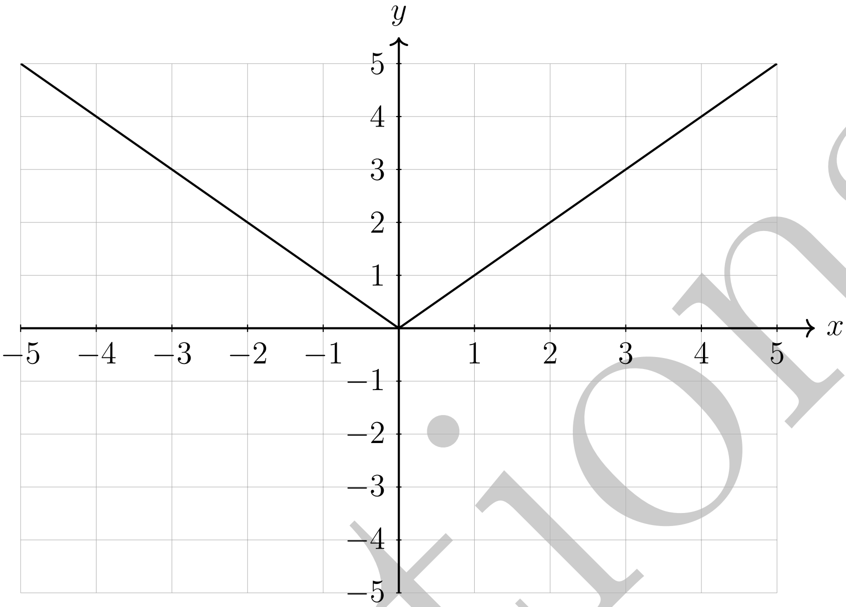
Foundation Mathematics 1017SCG
Week 8 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1.

$$y = |x|$$

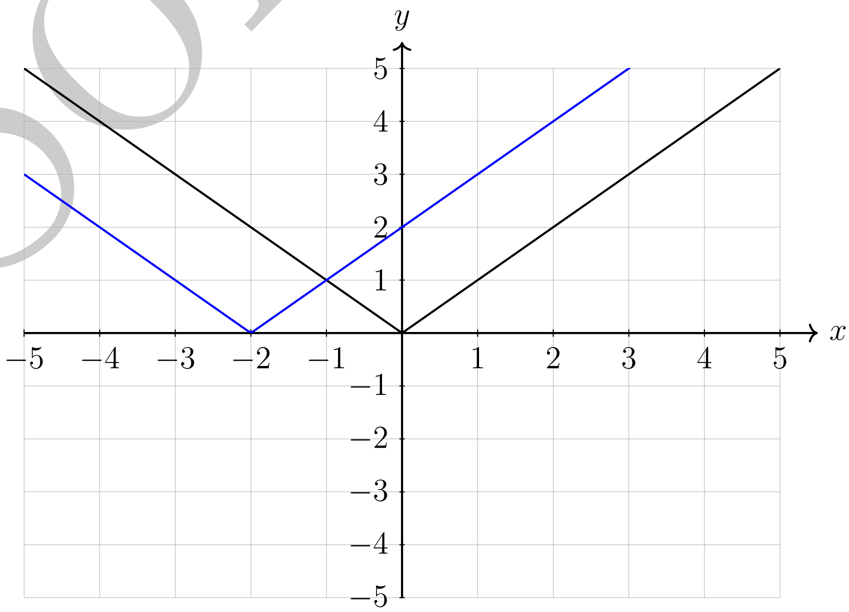


In the solutions for Question 1, the graph of $y = |x|$ is shown in black while the solution to the question is shown in blue.

2. (a)

$$y = |x + 2|$$

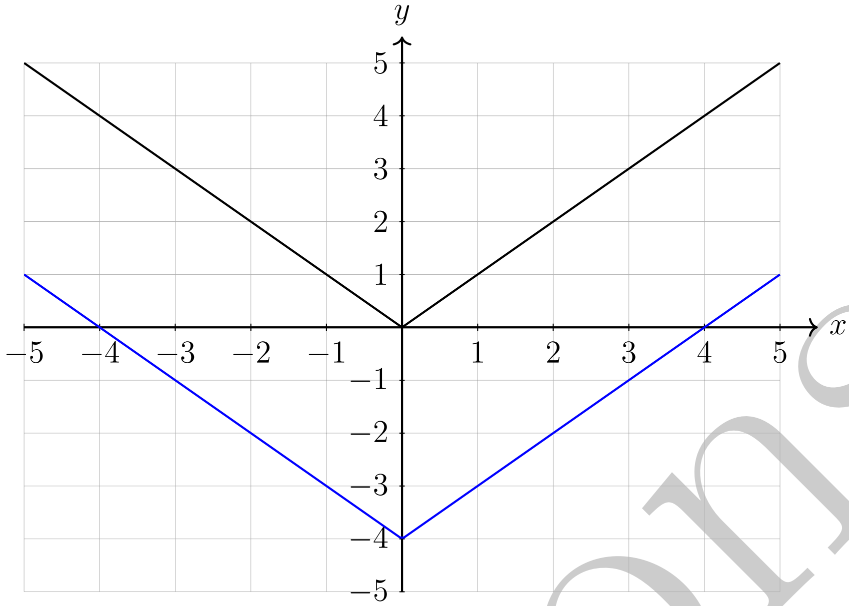
- Horizontal shift of 2 to the left



(b)

$$y = |x| - 4$$

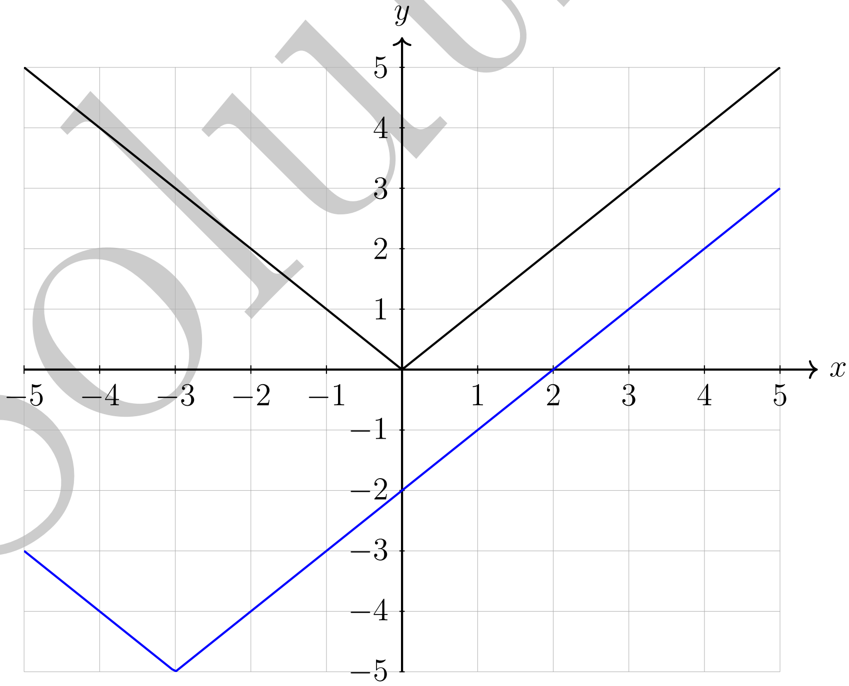
- Vertical shift of 4 down



(c)

$$y = |x + 3| - 5$$

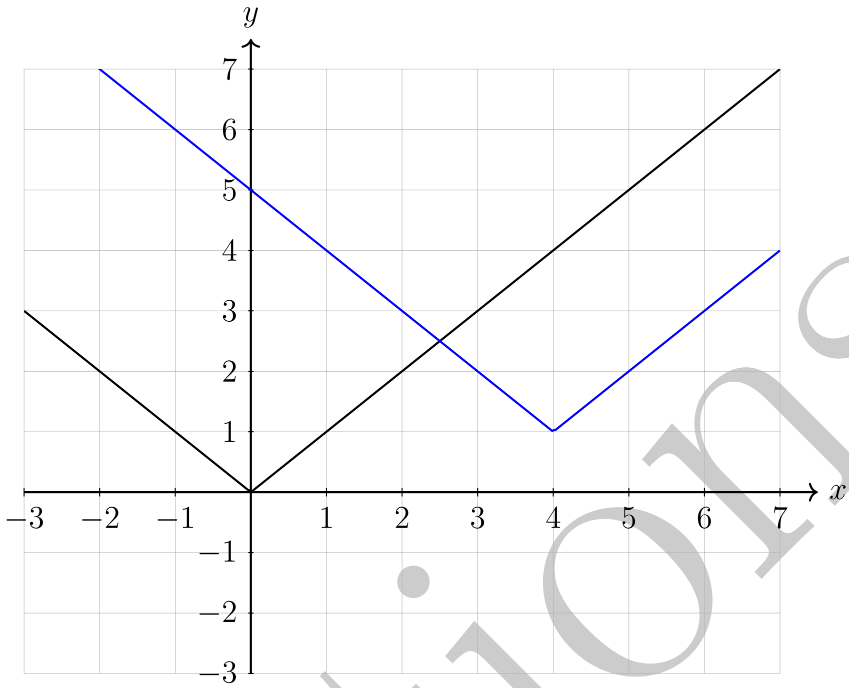
- Horizontal shift of 3 to the left
- Vertical shift of 5 down



(d)

$$y = |x-4| + 1$$

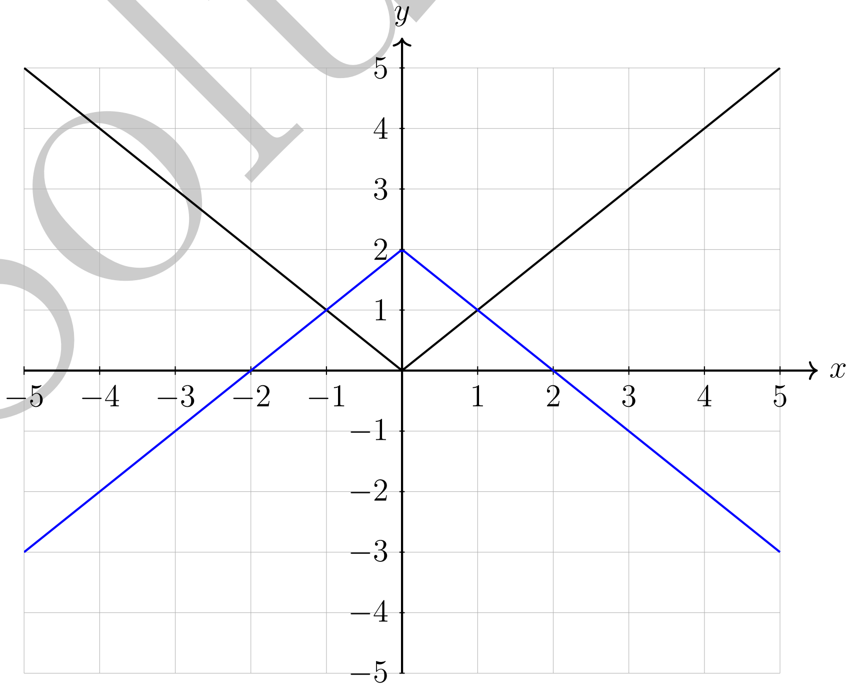
- Horizontal shift of 4 to the right
- Vertical shift of 1 up



(e)

$$y = -|x| + 2$$

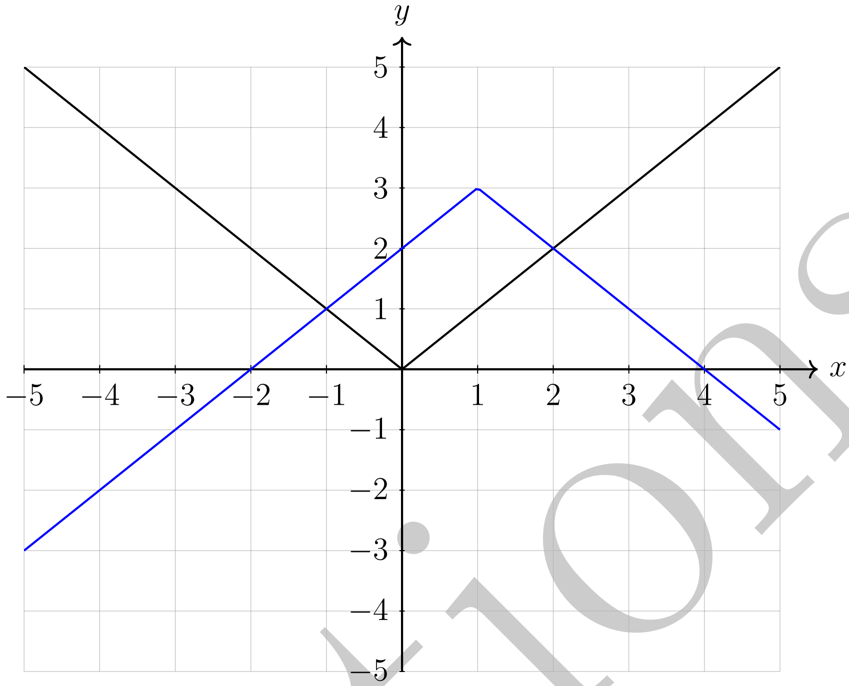
- Reflected across the x-axis
- Vertical shift of 2 up



(f)

$$y = -|x-1| + 3$$

- Horizontal shift of 1 to the right
- Reflected across the x-axis
- Vertical shift of 3 up



3.

$$f(x) = \begin{cases} 2x - 1 & x \geq 1 \\ -x + 4 & x < 1 \end{cases}$$

(a)

$$\begin{aligned} f(-1) &= -(-1) + 4 \\ &= 1 + 4 \\ &= 5 \end{aligned}$$

(b)

$$\begin{aligned} f(0) &= -(0) + 4 \\ &= 4 \end{aligned}$$

(c)

$$\begin{aligned} f(1) &= 2 \times (1) - 1 \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

(d)

$$\begin{aligned} f(2) &= 2 \times (2) - 1 \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

4.

$$g(x) = \begin{cases} 2-x & x \leq -2 \\ 5x+1 & -2 < x < 3 \\ 2x-4 & x \geq 3 \end{cases}$$

(a)

$$\begin{aligned} g(-3) &= 2-(-3) \\ &= 2+3 \\ &= 5 \end{aligned}$$

(b)

$$\begin{aligned} g(-2) &= 2-(-2) \\ &= 2+2 \\ &= 4 \end{aligned}$$

(c)

$$\begin{aligned} g(0) &= 5 \times (0) + 1 \\ &= 0+1 \\ &= 1 \end{aligned}$$

(d)

$$\begin{aligned} g(5) &= 2 \times (5) - 4 \\ &= 10-4 \\ &= 6 \end{aligned}$$

5.

$$h(x) = \begin{cases} x^2+4x & x < -5 \\ 4 & -5 \leq x < 2 \\ \sqrt{x} & x \geq 2 \end{cases}$$

(a)

$$h(-3) = 4$$

(b)

$$\begin{aligned} h(-6) &= (-6)^2 + 4 \times (-6) \\ &= 36-24 \\ &= 12 \end{aligned}$$

(c)

$$h(0) = 4$$

(d)

$$\begin{aligned} h(9) &= \sqrt{9} \\ &= 3 \end{aligned}$$

6. (a)

$$
\begin{gathered}
f(x) = 7x - 2 \\
\text{Let } y = 7x - 2 \\
y + 2 = 7x - 2 + 2 \\
y + 2 = 7x \\
\frac{y+2}{7} = \frac{7x}{7} \\
x = \frac{y+2}{7} \\
f^{-1}(y) = \frac{y+2}{7} \\
f^{-1}(x) = \frac{x+2}{7}
\end{gathered}
$$

(b)

$$
\begin{gathered}
f(x) = 3x + 4 \\
\text{Let } y = 3x + 4 \\
y - 4 = 3x + 4 - 4 \\
y - 4 = 3x \\
\frac{y-4}{3} = \frac{3x}{x} \\
x = \frac{y-4}{3} \\
f^{-1}(y) = \frac{y-4}{3} \\
f^{-1}(x) = \frac{x-4}{3}
\end{gathered}
$$

(c)

$$
\begin{gathered}
f(x) = \frac{2x-1}{5} \\
\text{Let } y = \frac{2x-1}{5} \\
y \times 5 = \frac{2x-1}{5} \times 5 \\
5y = 2x - 1 \\
5y + 1 = 2x - 1 + 1 \\
5y + 1 = 2x \\
\frac{5y+1}{2} = \frac{2x}{2} \\
x = \frac{5y+1}{2} \\
f^{-1}(y) = \frac{5y+1}{2} \\
f^{-1}(x) = \frac{5x+1}{2}
\end{gathered}
$$

(d)

$$f(x) = \frac{6x-7}{2}$$
$$\text{Let } y = \frac{6x-7}{2}$$
$$y \times 2 = \frac{6x-7}{2} \times 2$$
$$2y = 6x - 7$$
$$2y + 7 = 6x - 7 + 7$$
$$2y + 7 = 6x$$
$$\frac{2y+7}{6} = \frac{6x}{6}$$
$$x = \frac{2y+7}{6}$$
$$f^{-1}(y) = \frac{2y+7}{6}$$
$$f^{-1}(x) = \frac{2x+7}{6}$$

(e)

$$f(x) = 4e^{5x}$$
$$\text{Let } y = 4e^{5x}$$
$$\frac{y}{4} = \frac{4e^{5x}}{4}$$
$$\frac{y}{4} = e^{5x}$$
$$\ln\left(\frac{y}{4}\right) = \ln\left(e^{5x}\right)$$
$$\ln\left(\frac{y}{4}\right) = 5x \times \ln(e)$$
$$\ln\left(\frac{y}{4}\right) = 5x$$
$$\frac{\ln\left(\frac{y}{4}\right)}{5} = \frac{5x}{5}$$
$$x = \frac{\ln\left(\frac{y}{4}\right)}{5}$$
$$f^{-1}(y) = \frac{\ln\left(\frac{y}{4}\right)}{5}$$
$$f^{-1}(x) = \frac{\ln\left(\frac{x}{4}\right)}{5}$$

(f)

$$f(x) = 7e^{-x}$$
$$\text{Let } y = 7e^{-x}$$
$$\frac{y}{7} = \frac{7e^{-x}}{7}$$
$$\frac{y}{7} = e^{-x}$$
$$\ln\left(\frac{y}{7}\right) = \ln\left(e^{-x}\right)$$
$$\ln\left(\frac{y}{7}\right) = -x \times \ln(e)$$
$$\ln\left(\frac{y}{7}\right) = -1x$$
$$\frac{\ln\left(\frac{y}{7}\right)}{-1} = \frac{-1x}{-1}$$
$$x = -\ln\left(\frac{y}{7}\right)$$
$$f^{-1}(y) = -\ln\left(\frac{y}{7}\right)$$
$$f^{-1}(x) = -\ln\left(\frac{x}{7}\right)$$

Alternatively, the inverse function could be written as $f^{-1}(x) = \ln\left(\dfrac{7}{x}\right)$.

(g)

$$f(x) = 2e^{x-1}$$
$$\text{Let } y = 2e^{x-1}$$
$$\frac{y}{2} = \frac{2e^{x-1}}{2}$$
$$\frac{y}{2} = e^{x-1}$$
$$\ln\left(\frac{y}{2}\right) = \ln\left(e^{x-1}\right)$$
$$\ln\left(\frac{y}{2}\right) = (x-1) \times \ln(e)$$
$$\ln\left(\frac{y}{2}\right) = x - 1$$
$$\ln\left(\frac{y}{2}\right) + 1 = x - 1 + 1$$
$$x = \ln\left(\frac{y}{2}\right) + 1$$
$$f^{-1}(y) = \ln\left(\frac{y}{2}\right) + 1$$
$$f^{-1}(x) = \ln\left(\frac{x}{2}\right) + 1$$

(h)

$$
\begin{gathered}
f(x) = e^{3x+4} \\
\text{Let } y = e^{3x+4} \\
\ln(y) = \ln\left(e^{3x+4}\right) \\
\ln(y) = (3x+4) \times \ln(e) \\
\ln(y) = 3x+4 \\
\ln(y) - 4 = 3x + 4 - 4 \\
\ln(y) - 4 = 3x \\
\frac{\ln(y) - 4}{3} = \frac{3x}{3} \\
x = \frac{\ln(y) - 4}{3} \\
f^{-1}(y) = \frac{\ln(y) - 4}{3} \\
f^{-1}(x) = \frac{\ln(x) - 4}{3}
\end{gathered}
$$

7.

$$f(x) = 5x \text{ and } g(x) = 2x + 1$$

(a)

$$
\begin{aligned}
f(g(x)) &= f(2x+1) \\
&= 5(2x+1) \\
&= 10x + 5
\end{aligned}
$$

(b)

$$
\begin{aligned}
g(f(x)) &= g(5x) \\
&= 2(5x) + 1 \\
&= 10x + 1
\end{aligned}
$$

8.

$$f(x) = 3x - 4 \text{ and } g(x) = 7x + 2$$

(a)

$$
\begin{aligned}
f(g(x)) &= f(7x+2) \\
&= 3(7x+2) - 4 \\
&= 21x + 6 - 4 \\
&= 21x + 2
\end{aligned}
$$

(b)

$$
\begin{aligned}
g(f(x)) &= g(3x-4) \\
&= 7(3x-4) + 2 \\
&= 21x - 28 + 2 \\
&= 21x - 26
\end{aligned}
$$

9.

$$f(x) = 4x \text{ and } g(x) = 2\cos(x)$$

(a)

$$\begin{aligned} f(g(x)) &= f(2\cos(x)) \\ &= 4 \times (2\cos(x)) \\ &= 8\cos(x) \end{aligned}$$

(b)

$$\begin{aligned} g(f(x)) &= g(4x) \\ &= 2\cos(4x) \end{aligned}$$

10.
$$f(x) = 3x - 1 \text{ and } g(x) = 5\sin(2x)$$

(a)

$$\begin{aligned} f(g(x)) &= f(5\sin(2x)) \\ &= 3 \times (5\sin(2x)) - 1 \\ &= 15\sin(2x) - 1 \end{aligned}$$

(b)

$$\begin{aligned} g(f(x)) &= g(3x - 1) \\ &= 5\sin(2(3x - 1)) \\ &= 5\sin(6x - 2) \end{aligned}$$

11. The transformations that would have occurred to obtain the blue graph are

- Horizontal shift by 2 to the left
- Vertical shift by 3 down

Therefore the equation for the blue graph is $y = |x + 2| - 3$.

12. The transformations that would have occurred to obtain the blue graph are

- Horizontal shift by 1 to the right
- Reflected across the x-axis
- Vertical shift by 4 up

Therefore the equation for the blue graph is $y = -|x - 1| + 4$.

13. We will start by finding the equation of the line on the 'left' of the graph. Two points on this line are (0,-5) and (1,-6).

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-6 - -5}{1 - 0} \\ &= \frac{-1}{1} \\ &= -1 \end{aligned}$$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - -5 &= -1(x - 0) \\ y + 5 &= -1x \\ y + 5 - 5 &= -x - 5 \\ y &= -x - 5 \end{aligned}$$

We will now find an equation for the line on the 'right' of the graph. Two points on this line are (2,-2) and (3,0).

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{0 - -2}{3 - 2} \\ &= \frac{2}{1} \\ &= 2 \end{aligned}$$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - -2 &= 2(x - 2) \\ y + 2 &= 2x - 4 \\ y + 2 - 2 &= 2x - 4 - 2 \\ y &= 2x - 6 \end{aligned}$$

Therefore, the piecewise function for the graph is

$$y = \begin{cases} -x - 5 & x \leq 1 \\ 2x - 6 & x > 1 \end{cases}$$

14. We will start by finding the equation of the line on the 'left' of the graph. Two points on this line are (-4,0) and (-3,1).

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{1 - 0}{-3 - -4} \\ &= \frac{1}{1} \\ &= 1 \end{aligned}$$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 0 &= 1(x - -4) \\ y &= 1(x + 4) \\ y &= x + 4 \end{aligned}$$

We will now find an equation for the line on the 'right' of the graph. Two points on this line are (0,0) and (1,-2).

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-2 - 0}{1 - 0} \\ &= \frac{-2}{1} \\ &= -2 \end{aligned}$$

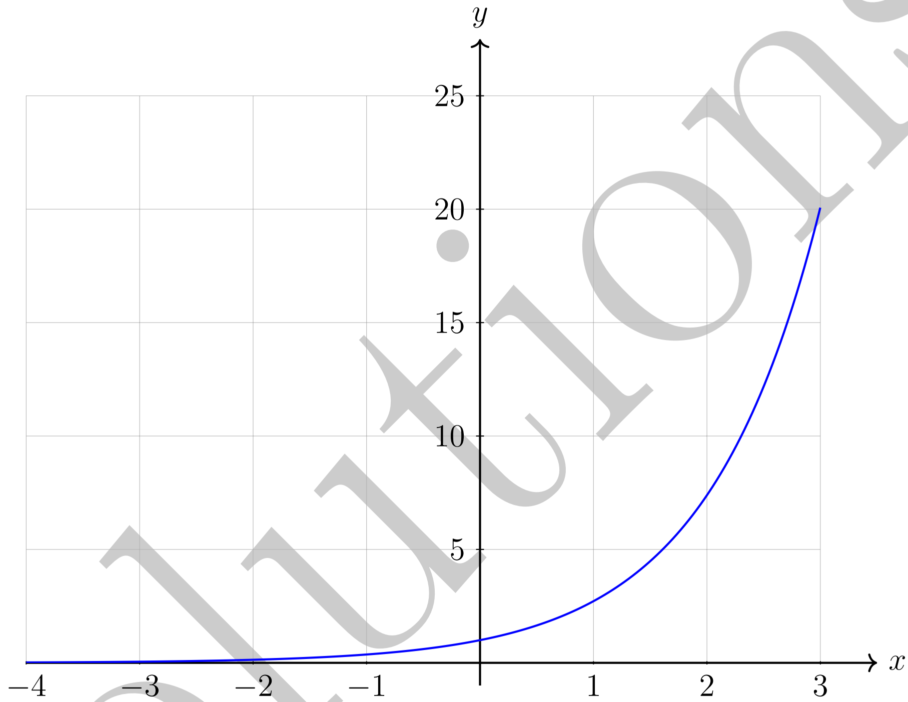
$$y - y_1 = m(x - x_1)$$
$$y - 0 = -2(x - 0)$$
$$y = -2x$$

Therefore, the piecewise function for the graph is

$$y = \begin{cases} x + 4 & x < -2 \\ -2x & x \geq -2 \end{cases}$$

15. (a) Table for $y = e^x$. Values for y are rounded to two decimal places where necessary.

x	-4	-3	-2	-1	0	1	2	3
y	0.02	0.05	0.14	0.37	1	2.72	7.39	20.09



(b) The table from part (a) has been extended to include, $x = -5$, $x = 4$ and $x = 5$.

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y	0.01	0.02	0.05	0.14	0.37	1	2.72	7.39	20.09	54.60	148.41

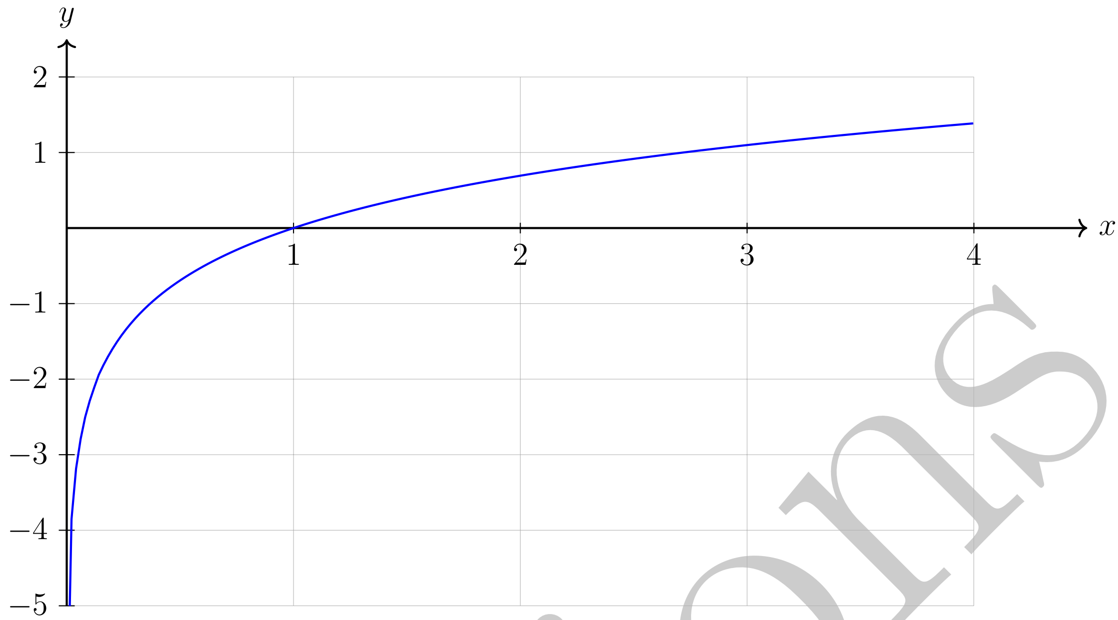
To accurately draw a graph of $y = e^x$ where $-5 \leq x \leq 5$, the y-axis would need values as low as 0.01 and as high as 148.41. Therefore a suitable y-axis could be $0 \leq y \leq 150$.

Note that for the domain given, there are several different ways we could construct our y-axis. If your y-axis can plot all points in the table above, it is most likely a suitable y-axis. For example, a slightly larger y-axis such that $-10 \leq y \leq 180$ would also be acceptable (although we should be careful not to go too far beyond 150 so that we are not 'wasting' space on the y-axis).

(c) Domain: All real values of x OR $-\infty < x < \infty$.
Range: All positive values of y OR $0 < y < \infty$. Note that $y = 0$ is NOT included in the range.

16. (a) Table for $y = \ln(x)$. Values for y are rounded to two decimal places where necessary.

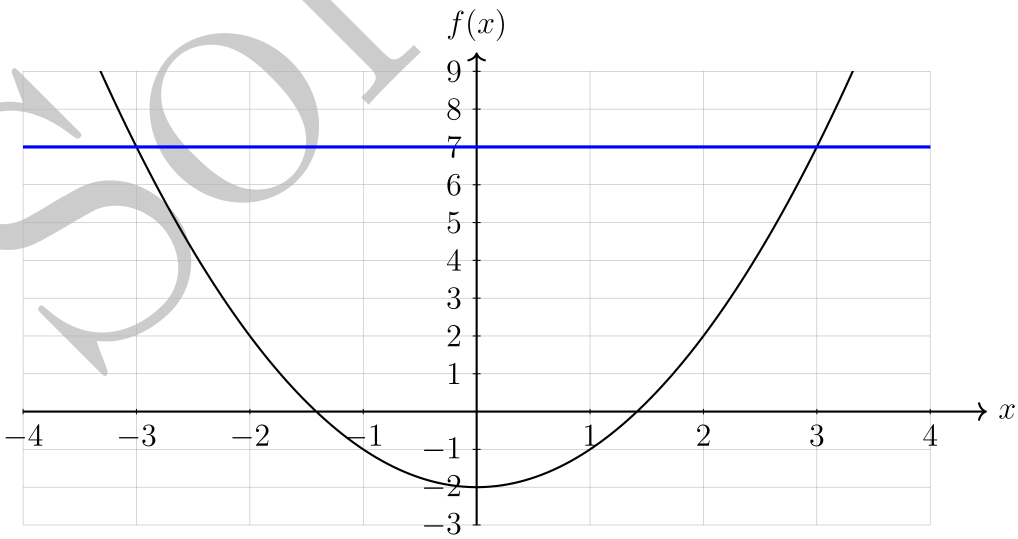
x	0.01	0.05	0.1	0.25	0.5	1	2	3	4
y	-4.61	-3.00	-2.30	-1.39	-0.69	0	0.69	1.10	1.39



(b) It is important that for the function $y = \ln(x)$, we only consider positive values of x as the function is not defined for negative values of x nor when x is equal to zero. (*In Foundation Mathematics, we only work within the real number system.*)

(c) Domain: $0 < x < \infty$ (Note that $x = 0$ is NOT included in the domain).
Range: All real values of y OR $-\infty < y < \infty$.

17. A graph of $f(x) = x^2 - 2$ is shown below. Note that the horizontal line (shown in blue) cuts the graph twice. Therefore, the function $f(x)$ fails the horizontal line test and does not have an inverse function.



Alternatively, consider the function $f(x) = x^2 - 2$. Consider that the function has given an output of 7 and we are trying to determine the input.

$$\begin{aligned} f(3) &= 3^2 - 2 \\ &= 9 - 2 \\ &= 7 \\ f(-3) &= (-3)^2 - 2 \\ &= 9 - 2 \\ &= 7 \end{aligned}$$

As multiple inputs give the same output, it is not possible to determine which input gave an output of 7. Therefore, $f(x)$ does not have an inverse function.

18.

$$\begin{aligned} f(x) &= \frac{x+1}{x-1} \\ \text{Let } y &= \frac{x+1}{x-1} \\ y(x-1) &= \frac{x+1}{x-1} \times (x-1) \\ y(x-1) &= x+1 \\ xy - y &= x + 1 \\ xy - x - y &= x - x + 1 \\ xy - x - y &= 1 \\ xy - x - y + y &= y + 1 \\ xy - x &= y + 1 \\ x(y-1) &= y + 1 \\ \frac{x(y-1)}{y-1} &= \frac{y+1}{y-1} \\ x &= \frac{y+1}{y-1} \\ f^{-1}(y) &= \frac{y+1}{y-1} \\ f^{-1}(x) &= \frac{x+1}{x-1} \end{aligned}$$